

Chi-square Test of Independence

Presentation Outline

1 Variables and Data

2 Test of Independence

Variables and Data Collection

- Variable 1
 - I categories
- Variable 2
 - J categories
- Data Collection and Summary
 - Random sample of size n from population
 - Cross-classify observations according to categories of two variables
 - Form into contingency table

Example - 3 x 4 Contingency Table

	Variable 2				
Variable 1	Cat. 1	Cat. 2	Cat. 3	Cat. 4	Total
Cat. 1	Y_{11}	Y_{12}	Y_{13}	Y_{14}	$Y_{1.}$
Cat. 2	Y_{21}	Y_{22}	Y_{23}	Y_{24}	$Y_{2.}$
Cat. 3	Y_{31}	Y_{32}	Y_{33}	Y_{34}	$Y_{3.}$
Total	$Y_{.1}$	$Y_{.2}$	$Y_{.3}$	$Y_{.4}$	n

Null and Alternative Hypotheses

- Definitions:
 - p_{ij} = population proportion for category i from Variable 1 and category j from Variable 2
 - $p_{i.}$ = population proportion for category i from Variable 1
 - $p_{.j}$ = population proportion for category j from Variable 2
- Null Hypothesis:
 - The two variables are independent in the population.
 - $H_0 : p_{ij} = p_{i.}p_{.j}$ for all $i = 1, 2, \dots, I$ and $j = 1, 2, \dots, J$
- Alternative Hypothesis:
 - The two variables are not independent in the population.
 - H_a : At least one $p_{ij} \neq p_{i.}p_{.j}$ for $i = 1, 2, \dots, I$ and $j = 1, 2, \dots, J$

Expected Values under H_0

If H_0 is true, then the expected value of each cell in the contingency table is:

$$E(Y_{ij}) = np_{ij} = np_{i.}p_{.j}$$

Estimate unknown probabilities $p_{i.}$ and $p_{.j}$ with our sample data:

$$\hat{p}_{i.} = \frac{Y_{i.}}{n} \quad \hat{p}_{.j} = \frac{Y_{.j}}{n}$$

Estimated expected value of each cell in the contingency table under H_0 :

$$\widehat{E(Y_{ij})} = n\hat{p}_{i.}\hat{p}_{.j} = n \left(\frac{Y_{i.}}{n} \right) \left(\frac{Y_{.j}}{n} \right) = \frac{Y_{i.}Y_{.j}}{n} = \frac{\text{row total} * \text{column total}}{\text{table total}}$$

Test Statistic

Test statistic is based on comparing $\widehat{E}(Y_{ij})$ to the observed values Y_{ij}

$$\chi^2 = \sum_{i=1}^I \sum_{j=1}^J \frac{(Y_{ij} - \widehat{E}(Y_{ij}))^2}{\widehat{E}(Y_{ij})}$$

- Sign of difference is not important - only the magnitude.
- Differences are weighted by the estimated expected value.

P-value

Assume $\widehat{E}(Y_{ij}) \geq 5$ for all $i = 1, 2, \dots, I$ and $j = 1, 2, \dots, J$

Distribution of X^2 is well-approximated by the Chi-square distribution with $(I - 1)(J - 1)$ degrees of freedom.

P-value:

$$P\left(\chi^2_{(I-1)(J-1)} > X^2\right)$$

Example - Test of Independence

A study from the 1970s involving more than 5000 students looked at the relationship between smoking habits of students and the smoking habits of their parents.

Variable 1 - Parent Smoking Status

- Categories: Neither Smokes, One Smokes, Both Smoke

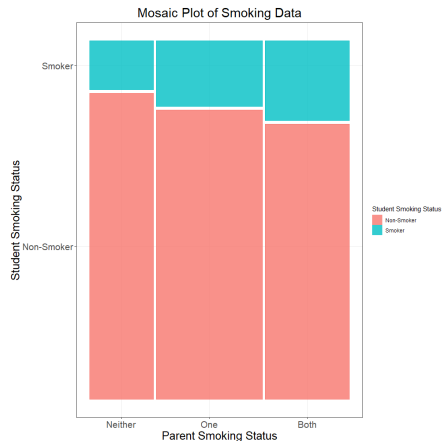
Variable 2 - Student Smoking Status

- Categories: Non-smoker, Smoker

Example - Test of Independence

Parent Status	Student Status		Total
	Non-smoker	Smoker	
Neither Smokes	1168	188	1356
One Smokes	1823	416	2239
Both Smoke	1380	400	1780
Total	4371	1004	5375

Example - Test of Independence



- Overall, we see a smaller proportion of students in the study are Smokers compared to Non-Smokers.
- The proportion of students who are Smokers is the lowest when Neither Parent Smokes and highest when Both Parents Smoke.

Example - Test of Independence

Null Hypothesis:

- H_0 : The smoking status of students and their parents are independent in this population.
- $H_0 : p_{ij} = p_{i.}p_{.j}$ for all $i = 1, 2, 3$ and $j = 1, 2$

Alternative Hypothesis:

- H_a : The smoking status of students and their parents are not independent in this population.
- H_a : at least one $p_{ij} \neq p_{i.}p_{.j}$ for $i = 1, 2, 3$ and $j = 1, 2$

Example - Test of Independence

Observed Data:

Parent Status	Student Status		Total
	Non-smoker	Smoker	
Neither Smokes	1168	188	1356
One Smokes	1823	416	2239
Both Smoke	1380	400	1780
Total	4371	1004	5375

Example - Test of Independence

Expected Values if H_0 is true:

Parent Status	Student Status		Total
	Non-smoker	Smoker	
Neither Smokes	1102.712	253.288	1356
One Smokes	1820.776	418.224	2239
Both Smoke	1447.512	332.488	1780
Total	4371	1004	5375

Example - Test of Independence

Test Statistic:

$$\begin{aligned} \chi^2 &= \frac{(1168 - 1102.712)^2}{1102.712} + \frac{(1823 - 1820.776)^2}{1820.776} + \dots \frac{(400 - 332.488)^2}{332.488} \\ &= 37.5663 \end{aligned}$$

Example - Test of Independence

P-value:

All estimated expected values are ≥ 5 , so we can use the $\chi^2_{(I-1)(J-1)} = \chi^2_{2*1} = \chi^2_2$ distribution to approximate the distribution of X^2 .

$$P(\chi^2_2 > 37.5663) < 0.0001$$

Conclusion:

We have extremely strong evidence that the smoking status of students is not independent from the smoking status of their parents in this population.